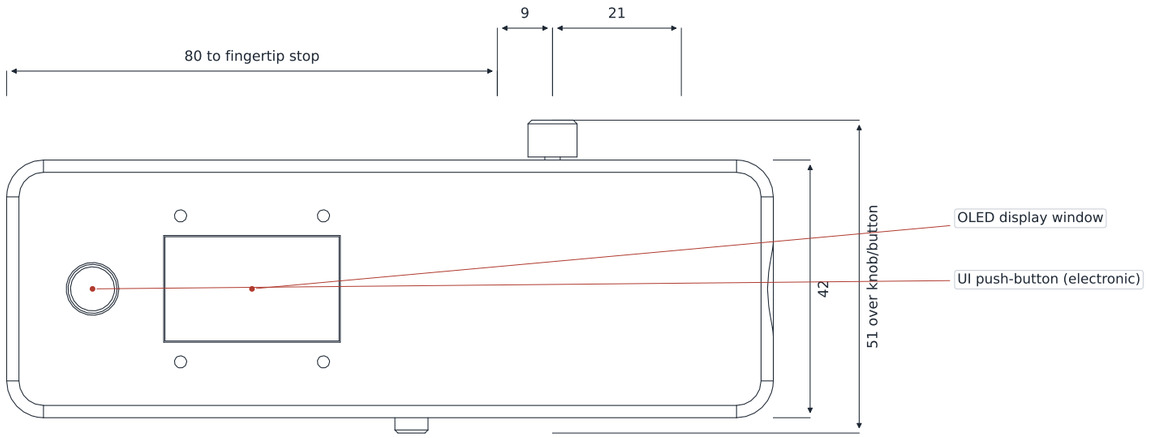
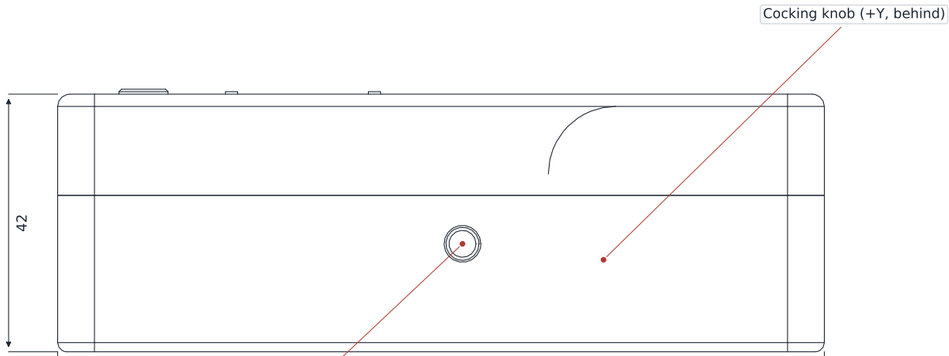
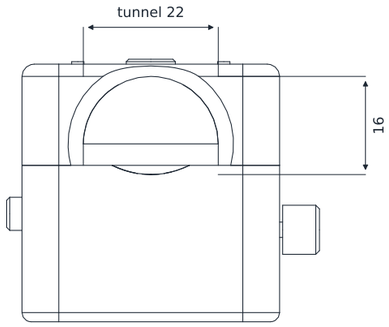



**PLAN (from +Z)**

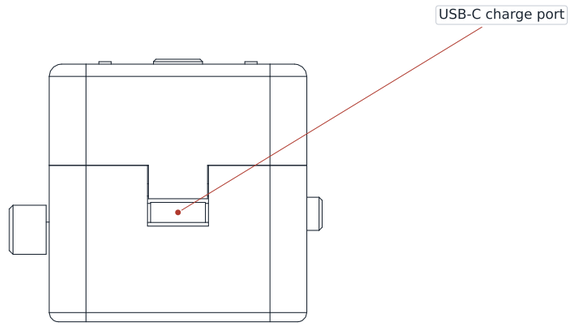



**SIDE ELEVATION (from -Y)**

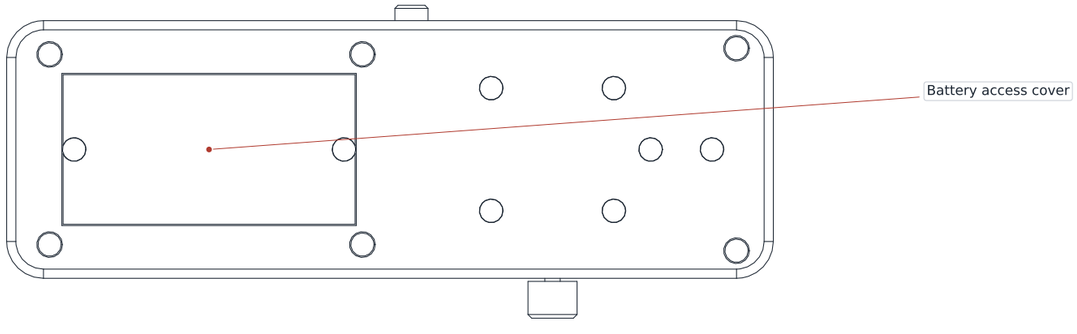
125 (housing)




**END VIEW - finger entry (+X)**




**END VIEW - rear, USB-C (-X)**



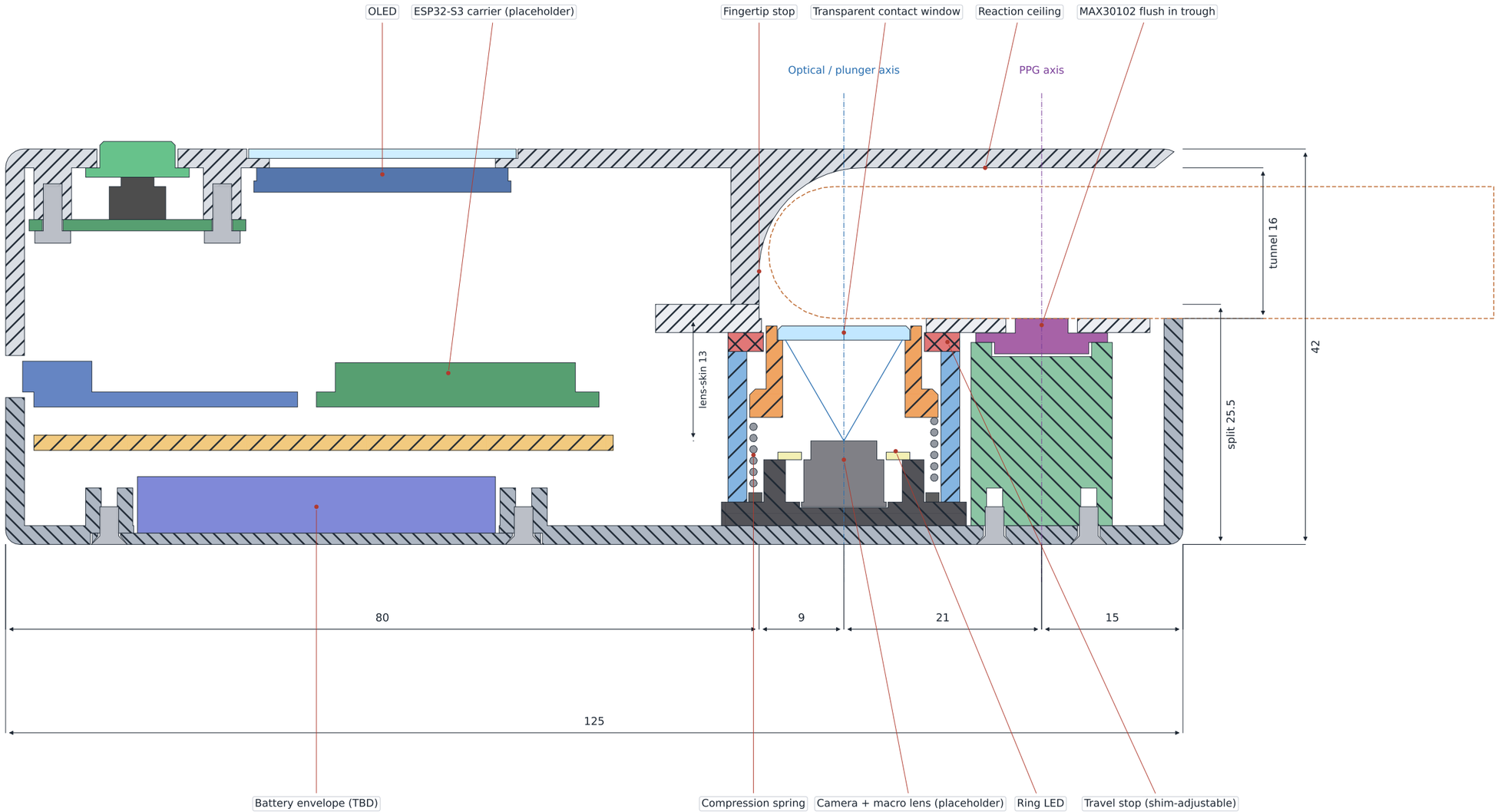

**UNDERSIDE (from -Z)**

**NOTES**

1. Housing envelope 125 x 42 x 42 mm measured from CAD shells; overall incl. knob / buttons 125 x 51 x 42.8 mm.
2. All dimensions are parametric (Params spreadsheet) and provisional.
3. Finger enters from +X, palm-side down; pulp rests in centring trough.
4. Split shell: lower = chassis/mechanism/electronics, upper = roof, display, UI button, finger tunnel and reaction ceiling. 6x M2x30 from below.
5. Mechanical release (-Y) is separate from the electronic UI button (top).
6. Tolerances: none specified. FDM prototype; sliding fits set by moving_part_clearance = 0.3 mm radial - tune by test print.
7. Materials are provisional assumptions; no part is validated for skin contact.
8. Force, contact pressure and dwell time NOT specified - load-cell calibration required.

**PROPOSED ENGINEERING CONCEPT - NOT VALIDATED - NOT FOR CLINICAL USE**

| ShockGuard - CRT / PPI fingertip triage device | |
|---|---|
| Assembly drawing - orthographic views, overall dimensions | Sheet 1 / 7  Rev A (concept) |
| Scale 1:1  Units: mm  Third-angle projection  2026-10-25 | Source: ShockGuard_Assembly.FCStd |




**NOTES**

Cocked state: spring at installed length, latch tooth over flange, window 0.8 mm below trough so the finger inserts without contact.

Release: spring drives plunger up to 4 mm; travel stop limits stroke. Max indentation if finger is already on the ceiling = 3.2 mm.

Re-cocking (pull knob down) lifts the window off the pulp - this is the refill-start event the camera times.

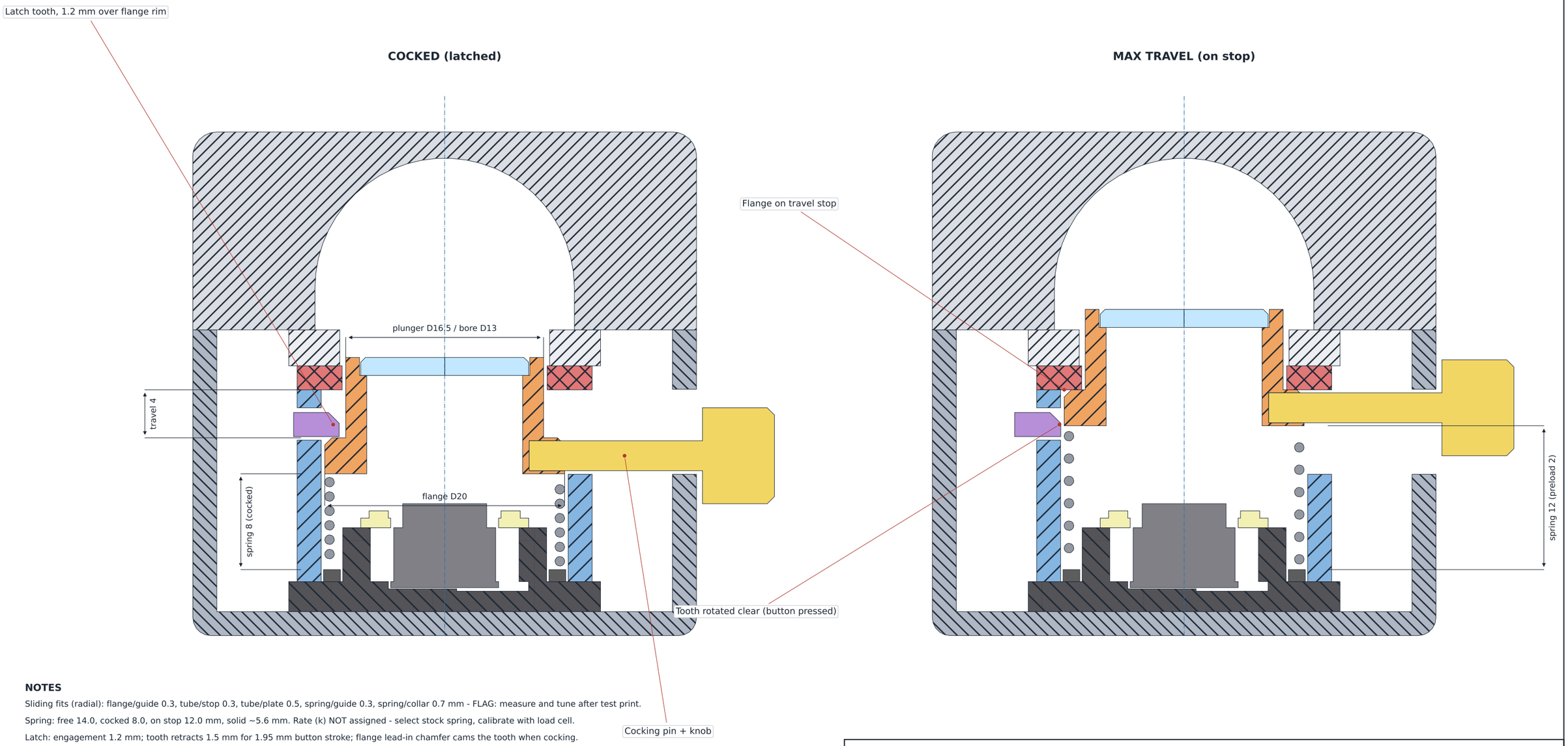
Camera views the pulp through the plunger bore (cone half-angle 30.1 deg, no obstruction found in CAD check). Macro / close-focus lens required.

Dashed outline = D14 mm reference fingertip (assumed, not anthropometric data).

Section hatching: printed housing parts. Component envelopes are placeholders.

**PROPOSED ENGINEERING CONCEPT - NOT VALIDATED - NOT FOR CLINICAL USE**

| **ShockGuard - CRT / PPI fingertip triage device** | |
|---|---|
| Section A-A (y = 0) - cocked, reference finger | Sheet 2 / 7  Rev A (concept) |
| Scale 2:1  Units: mm  Third-angle projection  2026-10-25 | Source: ShockGuard_Assembly.FCStd |




**NOTES**

Sliding fits (radial): flange/guide 0.3, tube/stop 0.3, tube/plate 0.5, spring/guide 0.3, spring/collar 0.7 mm - FLAG: measure and tune after test print.

Spring: free 14.0, cocked 8.0, on stop 12.0 mm, solid ~5.6 mm. Rate (k) NOT assigned - select stock spring, calibrate with load cell.

Latch: engagement 1.2 mm; tooth retracts 1.5 mm for 1.95 mm button stroke; flange lead-in chamfer cams the tooth when cocking.

CAD checks: latch blocks upward motion when cocked; travel stop blocks over-travel; no unintended interference in any state.

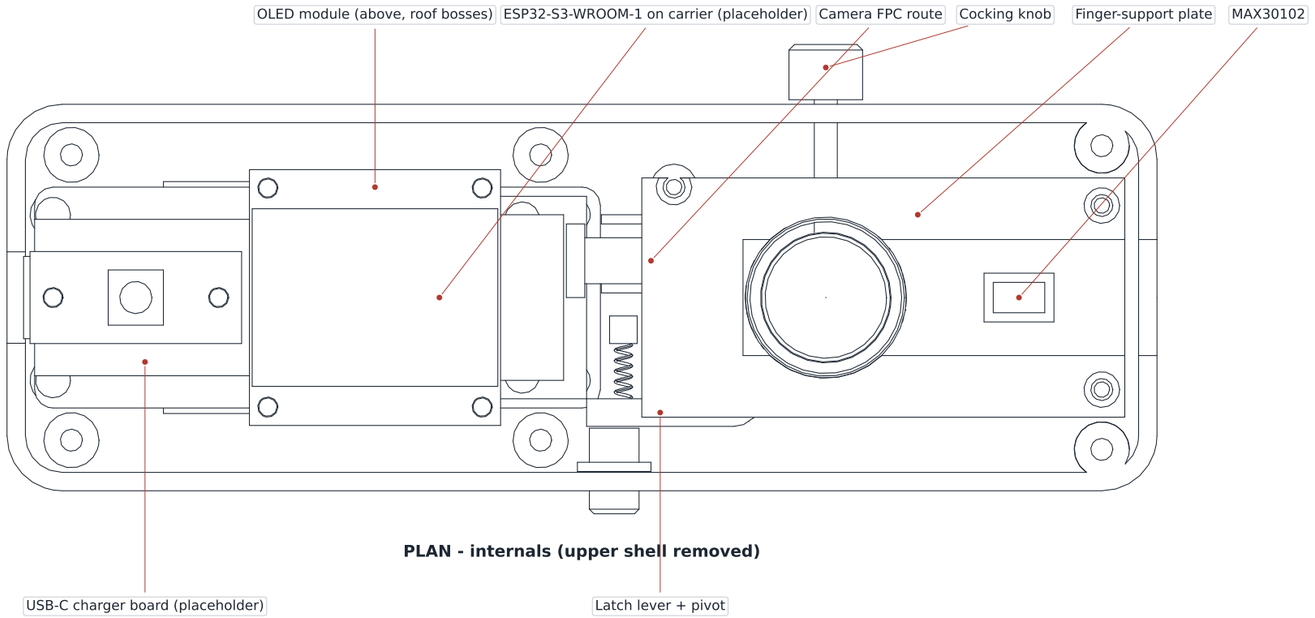
A fixed stroke does NOT guarantee a fixed force across fingers of different size / stiffness - see notes.

Review before human testing: pinch points at window/aperture and knob slot, accidental release, spring retention.

**PROPOSED ENGINEERING CONCEPT - NOT VALIDATED - NOT FOR CLINICAL USE**

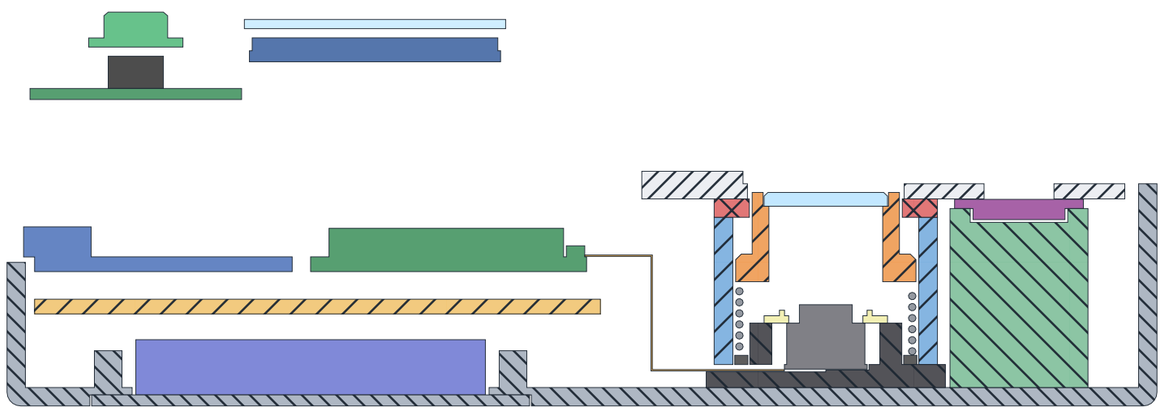
## ShockGuard - CRT / PPI fingertip triage device

| | |
|---|---|
| Section B-B - plunger mechanism, cocked / max travel | Sheet 3 / 7  Rev A (concept) |
| Scale 3:1  Units: mm  Third-angle projection  2026-10-25 | Source: ShockGuard_Assembly.FCStd |




**PLAN - internals (upper shell removed)**

**SECTION C-C (y = 2) - stack-up, upper shell omitted**



**NOTES**

Minimum clearance, moving parts -> electronics (all 4 states):

Release_Trigger -> ESP32_S3_Board: 0.4 mm

Compression_Spring -> Camera_FPC: 2.007 mm

Compression_Spring -> Ring_LED: 2.2 mm

Plunger_Body -> Ring_LED: 3.106 mm

Compression_Spring -> Camera_Module: 3.188 mm

Cocking_Feature -> Ring_LED: 3.433 mm

Battery: configurable envelope; chemistry, capacity and protection NOT selected.

Battery removable downward through floor door; tray lifts out with both PCBs

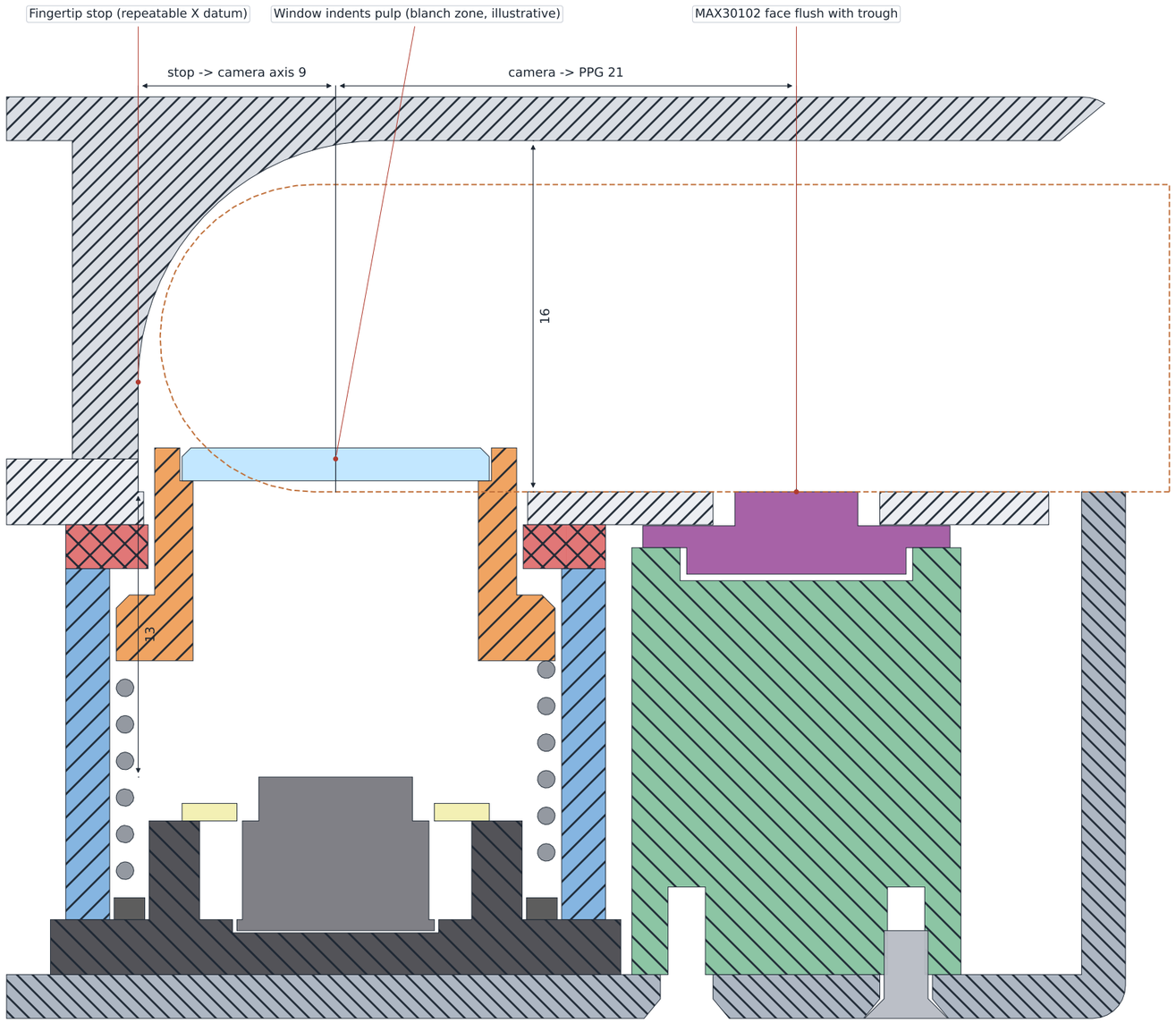
(USB-C slot is open-topped; upper-shell tongue closes it).

Board outlines are PLACEHOLDERS - replace with verified vendor drawings.

Keep WROOM antenna clear of screws/latch pivot (RF keep-out to be checked).

**PROPOSED ENGINEERING CONCEPT - NOT VALIDATED - NOT FOR CLINICAL USE**

| ShockGuard - CRT / PPI fingertip triage device | |
|---|---|
| Electronics compartment layout - plan with shells removed | Sheet 4 / 7  Rev A (concept) |
| Scale 1.5:1  Units: mm  Third-angle projection  2026-10-25 | Source: ShockGuard_Assembly.FCStd |




**NOTES**

The camera, ring LED and contact window share one axis, so the same pulp region is imaged every cycle (subject to optical validation).

PPG sits proximal to the blanch zone so it is not under the plunger; whether PPI is affected by distal compression must be tested.

Viewed skin diameter ~14.0 mm (limited by window D14). Window glare/reflection, colour calibration and focus are unverified.

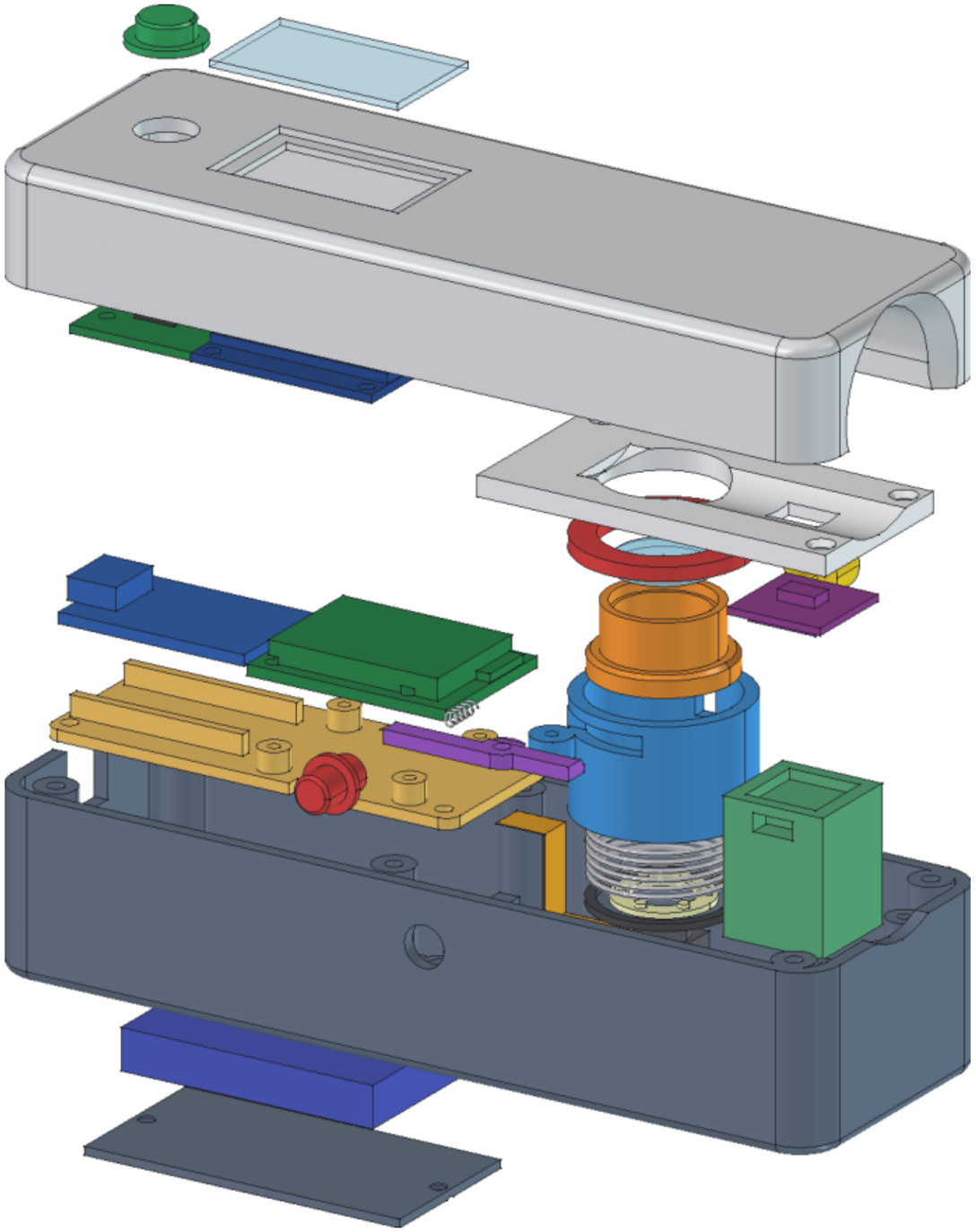
Indentation in this configuration is ILLUSTRATIVE (contact_indent_nominal) - real indentation depends on spring force and tissue stiffness.

Fingertip interface requires ergonomic and safety testing across finger sizes before any human use.

**PROPOSED ENGINEERING CONCEPT - NOT VALIDATED - NOT FOR CLINICAL USE**

| **ShockGuard - CRT / PPI fingertip triage device** | |
|---|---|
| Detail - measurement zone (released / contact) | Sheet 5 / 7  Rev A (concept) |
| Scale 4:1  Units: mm  Third-angle projection  2026-10-25 | Source: ShockGuard_Assembly.FCStd |

## EXPLODED VIEW (rendered from CAD, explode = 1)



## BILL OF MATERIALS

| ITEM | PART | QTY | MATERIAL / SELECTION | STATUS |
|---|---|---|---|---|
| 1 | Upper_Enclosure | 1 | PETG / ASA (FDM) - prototype only | ASSUMED |
| 2 | Lower_Enclosure | 1 | PETG / ASA (FDM) - prototype only | ASSUMED |
| 3 | Finger_Support | 1 | PETG (FDM) prototype; skin-contact material TBD. | VERIFY |
| 4 | Plunger_Body | 1 | Acetal (POM) or SLA resin; machined for repeata. | VERIFY |
| 5 | Contact_Pad | 1 | Optically clear PC / PMMA / glass (TBD; skin-co. | VERIFY |
| 6 | Compression_Spring | 1 | Music wire / 302 SS (stock spring TBD) | PLACEHOLDER |
| 7 | Spring_Seat | 1 | Acetal / printed PETG | ASSUMED |
| 8 | Guide | 1 | Acetal / PETG (bore reamed or sleeved) | VERIFY |
| 9 | Travel_Stop | 1 | Acetal / PETG; shim stack for stroke calibration | VERIFY |
| 10 | Cocking_Feature | 1 | Steel dowel pin pressed into printed knob | ASSUMED |
| 11 | Release_Trigger | 1 | Acetal / PETG (FDM fine layer) - fatigue test | VERIFY |
| 12 | Release_Button | 1 | PETG / ASA (FDM) | ASSUMED |
| 13 | Latch_Return_Spring | 1 | Stainless spring wire (stock, TBD) | PLACEHOLDER |
| 14 | Optical_Mount | 1 | PETG (FDM) / SLA black resin | ASSUMED |
| 15 | Camera_Module | 1 | OV2640-class camera + macro lens (TBD) | PLACEHOLDER |
| 16 | Ring_LED | 1 | Ring PCB with white/RGB LEDs (colour-consistent. | PLACEHOLDER |
| 17 | MAX30102_Module | 1 | MAX30102 breakout (TBD) | PLACEHOLDER |
| 18 | Sensor_Mount | 1 | PETG (FDM) | VERIFY |
| 19 | OLED_Module | 1 | 0.96in 128x64 I2C OLED (SSD1306-class, TBD) | PLACEHOLDER |
| 20 | Display_Window | 1 | Clear PMMA / PC 1 mm (laser cut) | ASSUMED |
| 21 | Electronics_Tray | 1 | PETG (FDM) | ASSUMED |
| 22 | ESP32_S3_Board | 1 | Custom carrier PCB + ESP32-S3-WROOM-1 (N16R8 su. | PLACEHOLDER |
| 23 | Charging_Board | 1 | 1S Li-ion charger + protection, USB-C (TBD) | PLACEHOLDER |
| 24 | Battery | 1 | 1S Li-ion/LiPo - chemistry, capacity, protectio. | PLACEHOLDER |
| 25 | Battery_Cover | 1 | PETG (FDM) | ASSUMED |
| 26 | Camera_FPC | 1 | Camera flex (length TBD by module) | PLACEHOLDER |
| 27 | Push_Button | 1 | PETG / TPU cap | ASSUMED |
| 28 | Tactile_Switch | 1 | 6x6 mm tactile switch (TBD) | PLACEHOLDER |
| 29 | Button_PCB | 1 | FR4 1.2 mm | PLACEHOLDER |
| 30 | Screw_M2x30_SHCS | 6 | Stainless A2 (assumed) | ASSUMED |
| 31 | Screw_M2x8_CSK_Top | 3 | Stainless A2 (assumed) | ASSUMED |
| 32 | Screw_M2x4_CSK_Bottom | 8 | Stainless A2 (assumed) | ASSUMED |
| 33 | Screw_M2x5_Pan | 15 | Stainless A2 (assumed) | ASSUMED |

**PROPOSED ENGINEERING CONCEPT - NOT VALIDATED - NOT FOR CLINICAL USE**

**ShockGuard - CRT / PPI fingertip triage device**

| | |
|---|---|
| Exploded assembly + bill of materials | Sheet 6 / 7  Rev A (concept) |
| Scale NTS  Units: mm  Third-angle projection  2026-10-25 | Source: ShockGuard_Assembly.FCStd |

| PARAMETER | VALUE | STATUS | PARAMETER | VALUE | STATUS | PARAMETER | VALUE | STATUS |
|---|---|---|---|---|---|---|---|---|
| overall_length | 125.0 | ASSUMED | cam_holder | 8.5 | PLACEHOLDER | explode | 0.0 | CONFIG |
| overall_width | 42.0 | ASSUMED | cam_holder_h | 4.5 | PLACEHOLDER | x_tip | 80.0 | DERIVED |
| overall_height | 42.0 | ASSUMED | cam_barrel_d | 7.0 | PLACEHOLDER | x_m | 89.0 | DERIVED |
| corner_radius | 6.0 | ASSUMED | cam_barrel_h | 2.0 | PLACEHOLDER | x_ppg | 110.0 | DERIVED |
| top_edge_fillet | 2.0 | ASSUMED | ring_light_diameter | 14.0 | PLACEHOLDER | z_mount_top | 4.5 | DERIVED |
| bottom_edge_fillet | 1.5 | ASSUMED | ring_light_inner | 9.0 | PLACEHOLDER | z_seat_top | 5.5 | DERIVED |
| enclosure_wall_thickness | 2.0 | VERIFY | ring_light_thickness | 0.8 | PLACEHOLDER | z_flange_bot | 13.5 | DERIVED |
| finger_opening_width | 22.0 | VERIFY | ring_led_radius | 5.5 | PLACEHOLDER | z_flange_top | 16.5 | DERIVED |
| finger_opening_height | 16.0 | VERIFY | max_pcb | 14.0 | PLACEHOLDER | z_guide_top | 20.5 | DERIVED |
| finger_support_length | 45.0 | ASSUMED | max_pcb_t | 1.0 | PLACEHOLDER | z_stop_top | 22.5 | DERIVED |
| trough_radius | 14.0 | ASSUMED | max_sensor_l | 5.6 | PLACEHOLDER | z_split | 25.5 | DERIVED |
| trough_depth | 1.5 | ASSUMED | max_sensor_w | 3.3 | PLACEHOLDER | z_fs | 24.0 | DERIVED |
| camera_mount_position | 9.0 | VERIFY | max_sensor_h | 1.55 | PLACEHOLDER | z_win_top | 23.2 | DERIVED |
| ppg_mount_position | 21.0 | VERIFY | oled_pcb_w | 27.3 | PLACEHOLDER | z_ceiling | 40.0 | DERIVED |
| plate_half_width | 13.0 | ASSUMED | oled_pcb_h | 27.8 | PLACEHOLDER | z_tunnel_axis | 29.0 | DERIVED |
| plunger_diameter | 16.5 | ASSUMED | oled_pcb_t | 1.2 | PLACEHOLDER | top_over_tunnel | 2.0 | DERIVED |
| plunger_bore | 13.0 | ASSUMED | oled_glass_w | 26.7 | PLACEHOLDER | z_lens_top | 11.0 | DERIVED |
| flange_diameter | 20.0 | ASSUMED | oled_glass_h | 19.3 | PLACEHOLDER | working_distance | 13.0 | DERIVED |
| flange_thickness | 3.0 | ASSUMED | oled_glass_t | 1.4 | PLACEHOLDER | z_pin | 15.0 | DERIVED |
| flange_chamfer | 0.6 | ASSUMED | oled_window_width | 24.0 | ASSUMED | guide_id | 20.6 | DERIVED |
| plunger_travel | 4.0 | VERIFY | oled_window_height | 13.0 | ASSUMED | guide_od | 24.6 | DERIVED |
| plunger_rest_gap | 0.8 | ASSUMED | oled_position_x | 40.0 | ASSUMED | stop_bore | 17.1 | DERIVED |
| contact_indent_nominal | 2.0 | VERIFY | oled_lens_t | 1.0 | ASSUMED | z_lever_bot | 16.6 | DERIVED |
| contact_pad_diameter | 14.0 | ASSUMED | button_position_x | 14.0 | ASSUMED | z_lever_top | 18.6 | DERIVED |
| contact_pad_thickness | 1.5 | ASSUMED | button_cap_d | 8.0 | ASSUMED | z_lever_mid | 17.6 | DERIVED |
| spring_outer_diameter | 20.0 | PLACEHOLDER | esp_x0 | 33.0 | ASSUMED | latch_pivot_x | 79.0 | DERIVED |
| spring_wire_diameter | 0.8 | PLACEHOLDER | esp_pcb_l | 30.0 | PLACEHOLDER | latch_pivot_y | -11.0 | DERIVED |
| spring_free_length | 14.0 | PLACEHOLDER | esp_pcb_w | 22.0 | PLACEHOLDER | latch_tooth_y | -8.8 | DERIVED |
| spring_installed_length | 8.0 | VERIFY | pcb_t | 1.6 | ASSUMED | release_button_x | 66.0 | DERIVED |
| spring_total_coils | 6.0 | PLACEHOLDER | wroom_l | 25.5 | PLACEHOLDER | latch_spring_x | 67.0 | DERIVED |
| guide_wall | 2.0 | ASSUMED | wroom_w | 18.0 | PLACEHOLDER | latch_post_y | -5.0 | DERIVED |
| spring_seat_thickness | 1.0 | ASSUMED | wroom_h | 3.1 | PLACEHOLDER | stroke | 0.0 | DERIVED |
| stop_thickness | 2.0 | VERIFY | chg_pcb_l | 28.0 | PLACEHOLDER | latch_angle | 0.0 deg | DERIVED |
| support_plate_thickness | 3.0 | ASSUMED | chg_pcb_w | 17.0 | PLACEHOLDER | spring_len_current | 8.0 | DERIVED |
| moving_part_clearance | 0.3 | VERIFY | usbc_w | 8.94 | PLACEHOLDER | latch_spring_len | 6.0 | DERIVED |
| plunger_hole_clearance | 0.5 | ASSUMED | usbc_l | 7.35 | PLACEHOLDER | z_tray_top | 11.6 | DERIVED |
| cocking_pin_diameter | 2.5 | ASSUMED | usbc_h | 3.26 | PLACEHOLDER | z_pcb_bot | 14.6 | DERIVED |
| knob_diameter | 8.0 | ASSUMED | battery_length | 38.0 | PLACEHOLDER | z_pcb_top | 16.2 | DERIVED |
| knob_length | 6.0 | ASSUMED | battery_width | 22.0 | PLACEHOLDER | z_usb_mid | 17.83 | DERIVED |
| latch_overlap | 1.2 | VERIFY | battery_thickness | 6.0 | PLACEHOLDER | z_oled_pcb_top | 38.6 | DERIVED |
| latch_thickness | 2.0 | ASSUMED | battery_x_center | 33.0 | ASSUMED | z_max_pcb_top | 22.45 | DERIVED |
| latch_release_angle | 8.6 deg | DERIVED | tray_z | 10.0 | ASSUMED | plate_x0 | 69.0 | DERIVED |
| release_button_x_offset | 23.0 | ASSUMED | tray_thickness | 1.6 | ASSUMED | plate_x1 | 121.5 | DERIVED |
| optical_mount_thickness | 2.5 | ASSUMED | standoff_h | 3.0 | ASSUMED | z_button_cap_bot | 39.0 | DERIVED |
| optical_mount_size | 26.0 | ASSUMED | electronics_clearance | 1.0 | VERIFY | z_switch_bot | 34.5 | DERIVED |
| collar_od | 17.0 | ASSUMED | screw_d | 2.0 | ASSUMED | hole_r | 1.2 | DERIVED |
| collar_id | 12.4 | ASSUMED | fastener_clearance | 0.2 | VERIFY | pilot_r | 0.85 | DERIVED |
| cam_board_size | 9.0 | PLACEHOLDER | pilot_d | 1.7 | VERIFY | spring_preload_at_stop | 2.0 | DERIVED |
| cam_board_t | 0.5 | PLACEHOLDER | state | 2.0 | CONFIG | | | |

VERIFY = must be confirmed by test/analysis;  PLACEHOLDER = replace with selected component datasheet;  ASSUMED = modelling assumption;  DERIVED = spreadsheet formula;  CONFIG = state switch.

**PROPOSED ENGINEERING CONCEPT - NOT VALIDATED - NOT FOR CLINICAL USE**

**ShockGuard - CRT / PPI fingertip triage device**

Master parameters (Params spreadsheet) - status flags

Sheet 7 / 7  Rev A (concept)

Scale -  Units: mm  Third-angle projection  2026-10-25

Source: ShockGuard_Assembly.FCStd